

261100690008

Prathiksha Patil

Part A: Authentication Log

Exercise 1: To print

Write sed commands to:

Print the first log entry.

sed -n '1p' auth.log

Print the fifth log entry.

sed -n '5p' auth.log

Print entries from lines 2 to 6.

sed -n '2,6p' auth.log

Print the last entry.

sed -n '\$p' auth.log

Print every second log entry starting from line 1.

sed -n '1~2p' auth.log

Print all entries containing Failed password.

sed -n '/Failed password/p' auth.log

Print all successful login entries containing Accepted password.

sed -n '/Accepted password/p' auth.log

Print entries from the first failed login to the authentication failure.

sed -n '/Failed password/,/authentication failure/p' auth.log

Exercise 2: Filtering Security Events

Write commands to:

Display failed login attempts.

sed -n '/Failed password/p' auth.log

Display successful login attempts.

sed -n '/Accepted password/p' auth.log

Display entries associated with the root account.

sed -n '/root/p' auth.log

Display entries containing Invalid user.

sed -n '/Invalid user/p' auth.log

Display all events related to IP address 172.16.0.8.

```
sed -n '/172.16.0.8/p' auth.log
```

Display all events that do not contain Accepted password.

```
sed -n '/Accepted password/!p' auth.log
```

Display all events except those generated by sudo.

```
sed -n '/sudo/!p' auth.log
```

Display only the entries that contain either Failed password or Invalid user using multiple -e commands

```
sed -n -e '/Failed password/p' -e '/Invalid user/p' auth.log
```

Exercise 3: Removing Irrelevant Entries

Without modifying the original file, display the log after:

Deleting successful login entries.

```
sed '/Accepted password/d' auth.log
```

Deleting entries containing Connection closed.

```
sed '/Connection closed/d' auth.log
```

Deleting the first three entries.

```
sed '1,3d' auth.log
```

Deleting entries from lines 4 to 7.

```
sed '4,7d' auth.log
```

Deleting all entries associated with 192.168.1.15.

```
sed '/192.168.1.15/d' auth.log
```

Deleting every second entry starting from the first line.

```
sed '1~2d' auth.log
```

Deleting all events that do not contain Failed password.

```
sed '/Failed password/!d' auth.log
```

Exercise 4: Classifying Security Events

Replace Accepted password with [SUCCESSFUL LOGIN].

```
sed 's/Accepted password/[SUCCESSFUL LOGIN]/g' auth.log
```

Replace Failed password with [FAILED LOGIN].

```
sed 's/Failed password/[FAILED LOGIN]/g' auth.log
```

Replace Invalid user with [INVALID USER].

```
sed 's/Invalid user/[INVALID USER]/g' auth.log
```

Replace authentication failure with [AUTHENTICATION FAILURE].
sed 's/authentication failure/[AUTHENTICATION FAILURE]/g' auth.log
Replace root with PRIVILEGED_USER.
sed 's/root/PRIVILEGED_USER/g' auth.log

Replace every occurrence of server with authentication-server.
sed 's/server/authentication-server/g' auth.log

Exercise 5: Masking Sensitive Information

Use sed to anonymise the log before sharing it.

Replace IP address 10.0.0.25 with X.X.X.X.
sed 's/10.0.0.25/X.X.X.X/g' auth.log

Replace 172.16.0.8 with SUSPICIOUS_IP.
sed 's/172.16.0.8/SUSPICIOUS_IP/g' auth.log

Replace username admin with USER_A.
sed 's/admin/USER_A/g' auth.log

Replace username student with USER_B.
sed 's/student/USER_B/g' auth.log

Replace all digits with X.
sed 's/[0-9]/X/g' auth.log

Replace periods in IP addresses with hyphens.
sed 's/\./-/g' auth.log

Replace all IP addresses in the sample file with [REDACTED_IP].
sed -E 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' auth.log

Part B: Web-Server Log

Use access.log to solve the below questions

Write sed commands to:

Print all requests that returned status code 200.
sed -n '/ 200\$/p' access.log

Print all requests that returned status code 401.
sed -n '/ 401\$/p' access.log

Print all requests that returned status code 403.
sed -n '/ 403\$/p' access.log

Print all requests to /admin/login.
sed -n '/admin/login/p' access.log
Print all POST requests.
sed -n '/POST/p' access.log

Print requests containing ../.
sed -n '/\.\.\./p' access.log

Print requests containing <script>.
sed -n '/<script>/p' access.log

Print requests containing OR.
sed -n '/OR/p' access.log

Print all requests except successful requests.
sed -n '/ 200\$/!p' access.log

Print all requests from 10.0.0.25.
sed -n '/^10.0.0.25/p' access.log

Exercise 7: Labelling Suspicious Web Requests

Use substitution to label the events:

Replace status code 200 with [ALLOWED].
sed 's/200/[ALLOWED]/g' access.log

Replace status code 401 with [UNAUTHORIZED].
sed 's/401/[UNAUTHORIZED]/g' access.log

Replace status code 403 with [BLOCKED].
sed 's/403/[BLOCKED]/g' access.log

Replace /admin/login with /RESTRICTED_RESOURCE.
sed 's/admin/login/RESTRICTED_RESOURCE/g' access.log

Replace GET with READ_REQUEST.
sed 's/GET/READ_REQUEST/g' access.log

Replace POST with SUBMIT_REQUEST.
sed 's/POST/SUBMIT_REQUEST/g' access.log

Exercise 8: Grouping Security Operations

Use { } to perform multiple commands.

For entries containing status code 401:
Replace 401 with [UNAUTHORIZED].
Print the modified line.

```
sed -n '/ 401$/s/401/[UNAUTHORIZED]/;p;}' access.log
```

For entries containing status code 403:

Replace 403 with [BLOCKED].

Replace the source IP address with [REDACTED_IP].

Print the modified entry.

```
sed -n '/ 403$/s/403/[BLOCKED]/; s/^[0-9.]+/[REDACTED_IP]/;p;}' access.log
```

```
sed -En '/ 403$/s/403/[BLOCKED]/; s/^[0-9.]+/[REDACTED_IP]/;p;}' access.log
```

For requests to /admin/login:

Replace the URL with /PROTECTED_PAGE.

Replace the source IP with [REDACTED_IP].

Print the result.

```
sed -n '/#/admin/login#{s#/admin/login#/PROTECTED_PAGE#; s/^[0-9.]+/[REDACTED_IP]/;p;}' access.log
```

Exercise 9: Adding Incident Labels

Insert heading before first line

```
sed '1i SECURITY AUTHENTICATION LOG' auth.log
```

Append message after final entry

```
sed '$a END OF SECURITY LOG' auth.log
```

Insert [WARNING] before every failed login

```
sed '/Failed password/i [WARNING]' auth.log
```

Append Review this event after every invalid-user entry

```
sed '/Invalid user/a Review this event' auth.log
```

Insert [SUSPICIOUS WEB REQUEST] before every 403 entry

```
sed '/ 403$/i [SUSPICIOUS WEB REQUEST]' access.log
```

Append [REQUIRES INVESTIGATION] after every 401 entry

```
sed '/ 401$/a [REQUIRES INVESTIGATION]' access.log
```

Replace Connection closed

```
sed 's/.*Connection closed.*/[CONNECTION TERMINATED]/' auth.log
```

Replace all successful web requests

```
sed '/ 200$/c [NORMAL WEB REQUEST]' access.log
```

Exercise 10: Separating Log Entries with G

Add blank line after every authentication event

sed 'G' auth.log

Add blank line after every failed login

sed '/Failed password/G' auth.log

Add blank line after every blocked web request

sed '/ 403\$/G' access.log

Add blank line after every second entry starting with first

sed '1~2G' auth.log

Exercise 11: Writing Selected Events to Files

Failed logins

sed -n '/Failed password/w failed_logins.txt' auth.log

Successful logins

sed -n '/Accepted password/w successful_logins.txt' auth.log

Invalid-user events

sed -n '/Invalid user/w invalid_users.txt' auth.log

401 web events

sed -n '/ 401\$/w unauthorized_requests.txt' access.log

403 web events

sed -n '/ 403\$/w blocked_requests.txt' access.log

Events related to 172.16.0.8

sed -n '/172.16.0.8/w suspicious_ip.txt' auth.log

Exercise 12: Combined Operations

Label successful and failed logins

sed -e 's/Accepted password/[SUCCESSFUL LOGIN]/g' -e 's/Failed password/[FAILED LOGIN]/g' auth.log

Label 401 and 403 web requests

sed -e 's/401/[UNAUTHORIZED]/g' -e 's/403/[BLOCKED]/g' access.log

Delete successful logins and print failed logins

sed '/Accepted password/d; /Failed password/p' auth.log

sed -n '/Accepted password/!{/Failed password/p;}' auth.log

Replace root and mask all IP addresses

```
sed -E 's/root/[PRIVILEGED_USER]/g; s/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' auth.log
```

Replace GET and POST

```
sed -e 's/GET/READ_REQUEST/g' -e 's/POST/SUBMIT_REQUEST/g' access.log
```

Exercise 13: security_report.sed

First create the file:

```
nano security_report.sed
```

Put these commands inside:

```
1i SECURITY INCIDENT REPORT
s/Failed password/[FAILED LOGIN]/g
s/Invalid user/[INVALID USER ATTEMPT]/g
s/Accepted password/[SUCCESSFUL LOGIN]/g
s/[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}/[REDACTED_IP]/g
$a END OF INCIDENT REPORT
```

```
sed -f security_report.sed auth.log
sed -f security_report.sed auth.log > incident_report.txt
cat incident_report.txt
```

Exercise 14: Modifying security.conf

```
nano security.conf
```

Enter:

```
PermitRootLogin yes
PasswordAuthentication yes
MaxAuthTries 6
FirewallStatus disabled
LogLevel INFO
GuestLogin enabled
```

Step 2: Make a backup

```
cp security.conf security.conf.backup
```

```
cat security.conf.backup
```

Step 3: Modify using sed -i

```
PermitRootLogin
```

```
sed -i 's/PermitRootLogin yes/PermitRootLogin no/' security.conf
```

```
MaxAuthTries
```

```
sed -i 's/MaxAuthTries 6/MaxAuthTries 3/' security.conf
```

FirewallStatus

```
sed -i 's/FirewallStatus disabled/FirewallStatus enabled/' security.conf
```

LogLevel

```
sed -i 's/LogLevel INFO/LogLevel VERBOSE/' security.conf
```

GuestLogin

```
sed -i 's/GuestLogin enabled/GuestLogin disabled/' security.conf
```

```
cat security.conf
```

PermitRootLogin no

PasswordAuthentication yes

MaxAuthTries 3

FirewallStatus enabled

LogLevel VERBOSE

GuestLogin disabled

Safer backup form

Instead of modifying directly:

```
sed -i.bak 's/PermitRootLogin yes/PermitRootLogin no/' security.conf
```

This automatically creates:

security.conf.bak

Final Bash Script

Create:

```
nano security_log_analyser.sh
```

```
#!/bin/bash
```

```
if [ $# -eq 0 ]
```

```
then
```

```
    echo "Error: Please provide a log filename."
```

```
    exit 1
```

```
fi
```

```
if [ ! -f "$1" ]
```

```
then
```

```
    echo "Error: File does not exist."
```

```
    exit 1
```

```
fi
```

```
logfile="$1"
```

```
echo "SECURITY ALERT REPORT" > security_alert_report.txt  
echo "===== " >> security_alert_report.txt
```

```
echo "" >> security_alert_report.txt  
echo "Failed Login Attempts:" >> security_alert_report.txt  
sed -n '/Failed password/p' "$logfile" | sed -E  
's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g; s/^/[ALERT] /' >> security_alert_report.txt
```

```
echo "" >> security_alert_report.txt  
echo "Invalid User Attempts:" >> security_alert_report.txt  
sed -n '/Invalid user/p' "$logfile" | sed -E 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g;  
s/^/[ALERT] /' >> security_alert_report.txt
```

```
echo "" >> security_alert_report.txt  
echo "END OF REPORT" >> security_alert_report.txt
```

```
cat security_alert_report.txt
```

```
chmod +x security_log_analyser.sh  
./security_log_analyser.sh auth.log
```

```
Security_alert_report.txt  
cat security_alert_report.txt
```